9.9 The Birthday Problem

WORKSHEET

The question

STEP II 2007, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q13

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Why is it easier to calculate the probability that *no* two guests share a birthday? Line the guests up: what is the chance that the second avoids the first's birthday? That the third avoids both?

STAGE 1 A useful approximation

Ask yourself What is the tangent line to $y = e^{-x}$ at $x = 0$?

STAGE 2 The probability of a shared birthday

Ask yourself Look at the exponent: $\frac{k(k-1)}{730} = \binom{k}{2} \Big/ 365$. What does that say in words?

STAGE 4 **Sharing with the host**

Ask yourself For each guest, what is the probability of *not* sharing the host's birthday? Are the guests independent of each other in this respect?

What's really going on

The square-root law

With n equally likely days instead of 365, the same argument gives

$$\mathrm{P}(\text{some match among } k) \approx 1 - e^{-k^2/(2n)},$$

so matches become likely once k is around $\sqrt{n}$, not n. More precisely, the probability reaches $\frac{1}{2}$ at $k \approx 1.18\sqrt{n}$, and the expected number of people you need to meet before the first repeat is about $\sqrt{\pi n/2}$ (for $n = 365$, about 24.6). This *square-root law* is the reason coincidences are so common: in any large collection of events, some pair is likely to match in some way, even though any particular match is unlikely.

Birthday attacks

Computers check that a file has not been tampered with by computing a *hash*: a fingerprint of, say, $n = 2^{128}$ possible values. A forger who wants two documents with the same hash (a harmless contract and a fraudulent one) does not need to try 2^{128} documents. By the birthday argument, generating about $\sqrt{2^{128}} = 2^{64}$ variants of each will probably produce a match. This *birthday attack* is why modern hash functions use 256 bits or more: half the bits are lost to the square-root law. Similar calculations decide how long random identifiers must be so that two computers never accidentally choose the same one.

Who first asked?

The birthday problem is usually credited to Richard von Mises, who discussed it in 1939, though it circulated earlier. Harold Davenport is also said to have known it earlier, and not to have published it because it seemed too simple. As Warren Weaver told the story, he liked to demonstrate it at dinners of army officers during the Second World War: he bet that two of the twenty-two people present shared a birthday, and

lost, until a waitress revealed that she shared her birthday with one of the officers.

Try this next

1 **The first repeat.** Try STEP II 2014, Question 13 (step.maths.org/questions/14-s2-q13). It studies the first repeated value from a random number generator, which is the birthday problem in another guise, and turns the probabilities into two surprising identities.

2 **Three in common.** Roughly how many people are needed for a better-than-even chance that some *three* share a birthday? (*Hint:* the expected number of triples sharing a birthday is $\binom{k}{3}/365^2$. Set this to about $\ln 2$.) The true answer is 88.

3 **Near misses.** How many people are needed for an even chance that two have birthdays on the same day or on adjacent days?

4 **Uneven birthdays.** Real birthdays are not uniform. Show that if the days have probabilities $p_1, \ldots, p_{365}$, the chance that two people share a birthday is $\sum p_i^2$, and that this is smallest when all p_i are equal. (So non-uniformity only makes matches more likely.)

Further reading

- **William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1** (Wiley), Chapter II. The birthday problem and its approximations.
- **Persi Diaconis & Frederick Mosteller, "Methods for studying coincidences"**, *Journal of the American Statistical Association* 84 (1989). Why coincidences are common; very readable.
- **David Hand, *The Improbability Principle*** (Bantam). The square-root law and other reasons why unlikely things happen all the time.

The full guided solution is in Section 9.9:
https://beyondstep.keithmcnulty.org/ch09/birthday.html